# Force Analysis (Ch. 11)

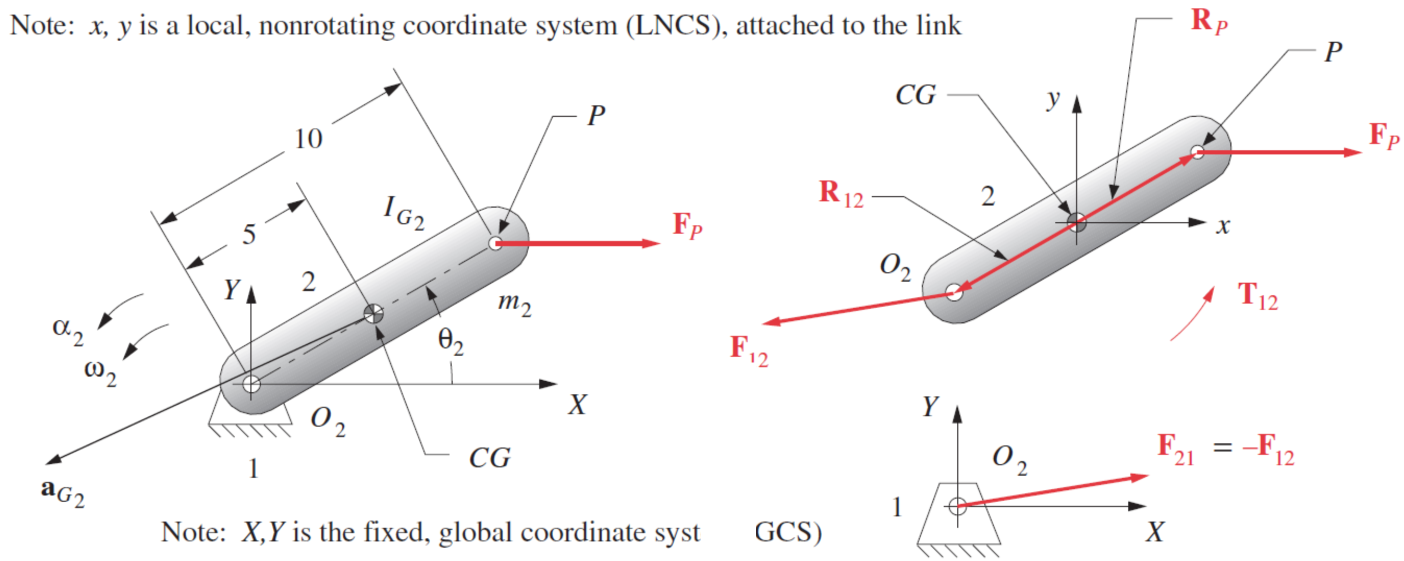
- **Newtonian Solution Method (FBD)**

$$\sum \mathbf{F} = m\mathbf{a} \qquad \sum \mathbf{T} = I_G \alpha$$

$$\sum F_x = m a_x \qquad \sum F_y = m a_y \qquad \sum T = I_G \alpha$$

**Note:** *Gravitational force is not considered as usually the accelerations of the COGs of the links are much higher than the gravitational acceleration. If this is not the case, gravitational force can be treated as an external force*.

## – Single link (pure rotation)




(*a*) Kinematic diagram

(*b*) Force (free-body) diagrams

$$F_{P_x} + F_{12_x} = m_2 a_{G_x}$$

$$F_{P_y} + F_{12_y} = m_2 a_{G_y}$$

$$T_{12} + \left(R_{12_x} F_{12_y} - R_{12_y} F_{12_x}\right) + \left(R_{P_x} F_{P_y} - R_{P_y} F_{P_x}\right) = I_G \alpha$$

$$\sum \mathbf{F} = \mathbf{F}_P + \mathbf{F}_{12} = m_2 \mathbf{a}_G$$

$$\sum \mathbf{T} = \mathbf{T}_{12} + \left(\mathbf{R}_{12} \times \mathbf{F}_{12}\right) + \left(\mathbf{R}_P \times \mathbf{F}_P\right) = I_G \alpha$$

$$\mathbf{A}\,\mathbf{B} = \mathbf{C}$$

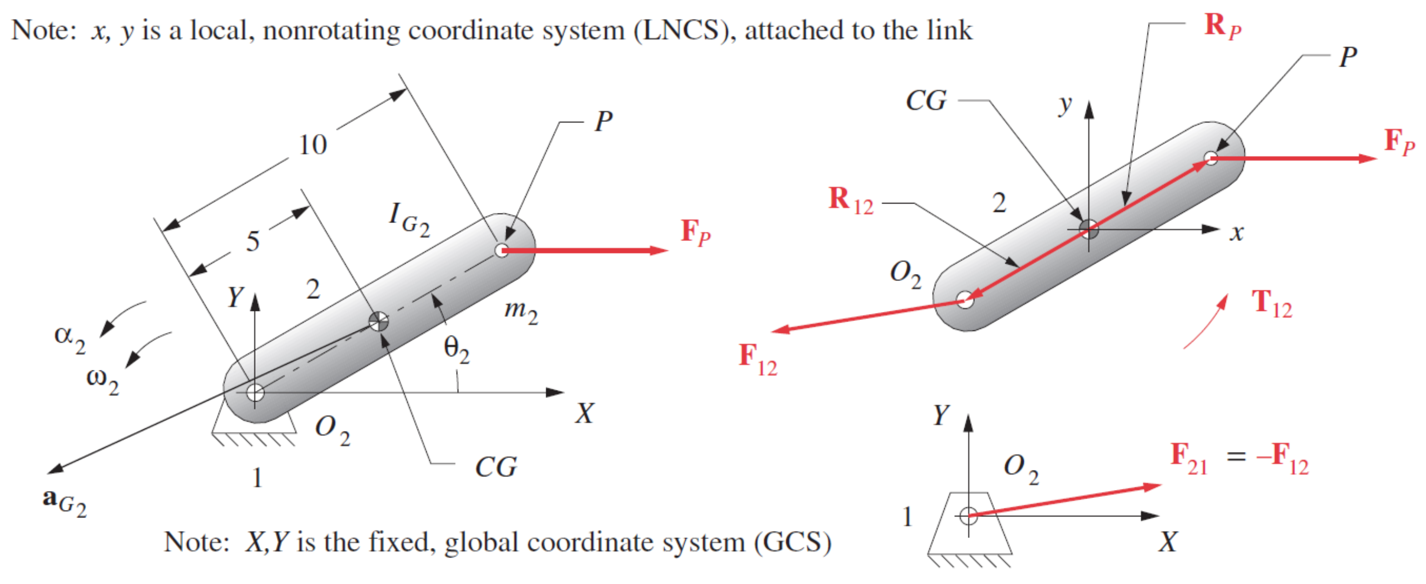
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -R_{12_y} & R_{12_x} & 1 \end{bmatrix} \begin{bmatrix} F_{12_x} \\ F_{12_y} \\ T_{12} \end{bmatrix} = \begin{bmatrix} m_2 a_{G_x} - F_{P_x} \\ m_2 a_{G_y} - F_{P_y} \\ I_G \alpha - \left(R_{P_x} F_{P_y} - R_{P_y} F_{P_x}\right) \end{bmatrix}$$

# Example 1

The 10-in-long link shown weighs 4 lb. Its *CG* is on the line of centers at the 5-in point. Its mass moment of inertia about its *CG* is 0.08 lb-in-sec$^2$. Its kinematic data are:

| $\theta_2$ deg | $\omega_2$ rad/sec | $\alpha_2$ rad/sec$^2$ | $a_{G_2}$ in/sec$^2$ |
|---|---|---|---|
| 30 | 20 | 15 | 2001 @ 208° |

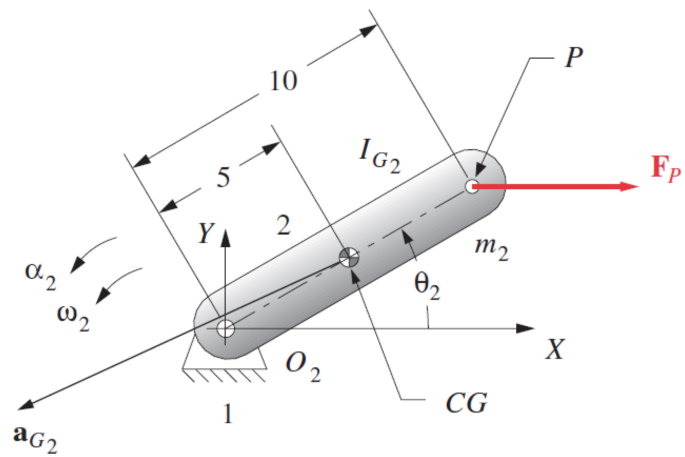
Find the <u>force **F**12</u> at pin joint *O*2 and the driving <u>torque **T**12</u> needed to maintain motion with the given acceleration for this instantaneous position of the link.

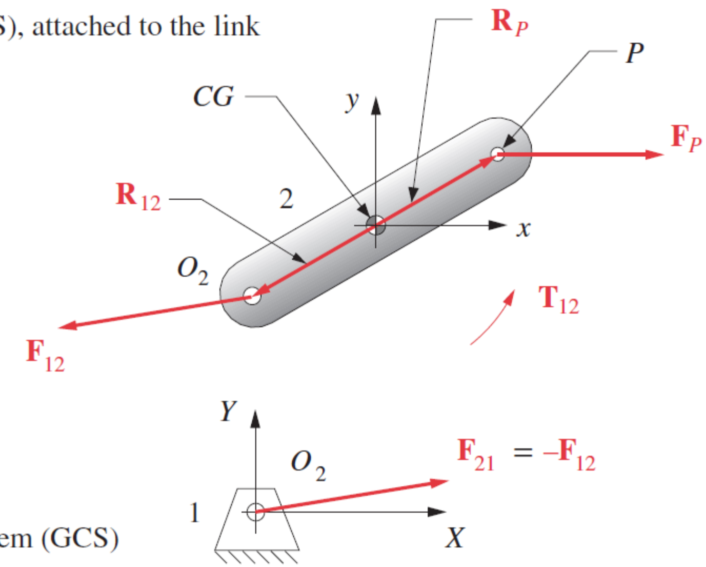



(*a*) Kinematic diagram (*b*) Force (free-body) diagrams

# Solution

Note: $x$, $y$ is a local, nonrotating coordinate system (LNCS), attached to the link




Note: $X,Y$ is the fixed, global coordinate system (GCS)

(*a*) Kinematic diagram

(*b*) Force (free-body) diagrams

$$m_2 = \frac{weight}{g} = \frac{4 \text{ lb}}{386 \text{ in/sec}^2} = 0.0104 \text{ blobs}$$

$\mathbf{R}_{12} = 5 \text{ in} @ \angle 210°;$ $\quad R_{12_x} = -4.33,$ $\quad R_{12_y} = -2.50$

$\mathbf{R}_P = - \mathbf{R}_{12}$ $\quad R_{P_x} = +4.33,$ $\quad R_{P_y} = +2.50$

$\mathbf{a}_G = 2001 @ \angle 208°;$ $\quad a_{G_x} = -1766.78,$ $\quad a_{G_y} = -939.41$

$\mathbf{F}_P = 40 @ \angle 0°;$ $\quad F_{P_x} = 40,$ $\quad F_{P_y} = 0$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -R_{12_y} & R_{12_x} & 1 \end{bmatrix} \begin{bmatrix} F_{12_x} \\ F_{12_y} \\ T_{12} \end{bmatrix} = \begin{bmatrix} m_2 a_{G_x} - F_{P_x} \\ m_2 a_{G_y} - F_{P_y} \\ I_G \alpha - \left( R_{P_x} F_{P_y} - R_{P_y} F_{P_x} \right) \end{bmatrix}$$

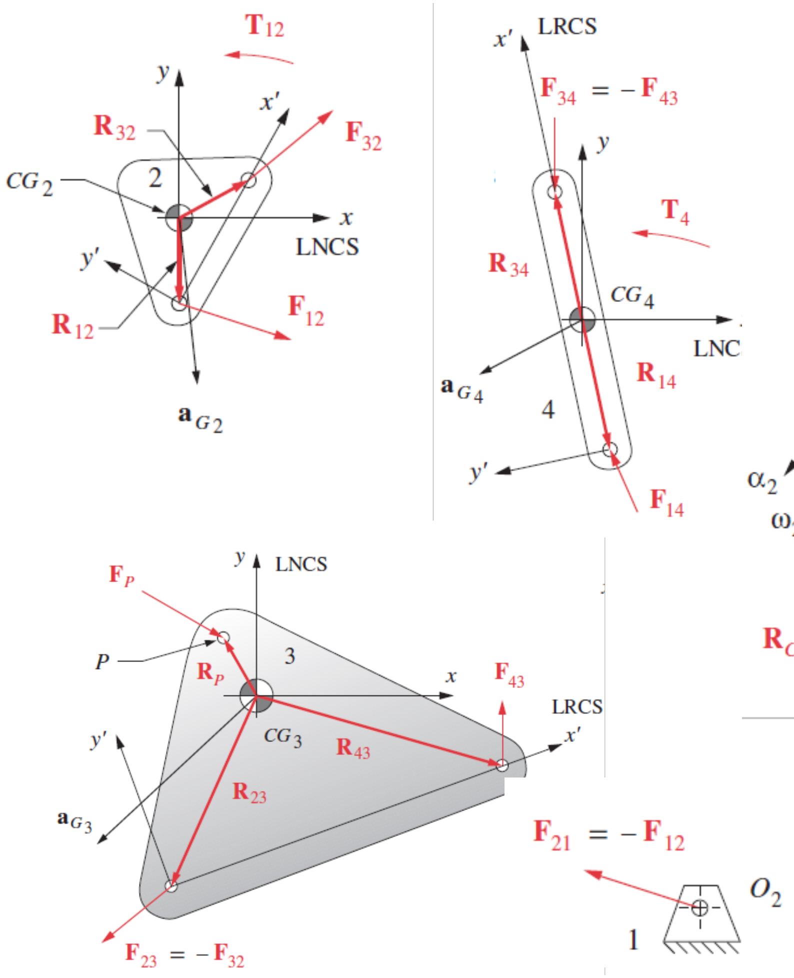
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2.50 & -4.33 & 1 \end{bmatrix} \begin{bmatrix} F_{12_x} \\ F_{12_y} \\ T_{12} \end{bmatrix} = \begin{bmatrix} (0.01)(-1766.78) - 40 \\ (0.01)(-939.41) - 0 \\ (0.08)(15) - \{(4.33)(0) - (2.5)(40)\} \end{bmatrix}$$

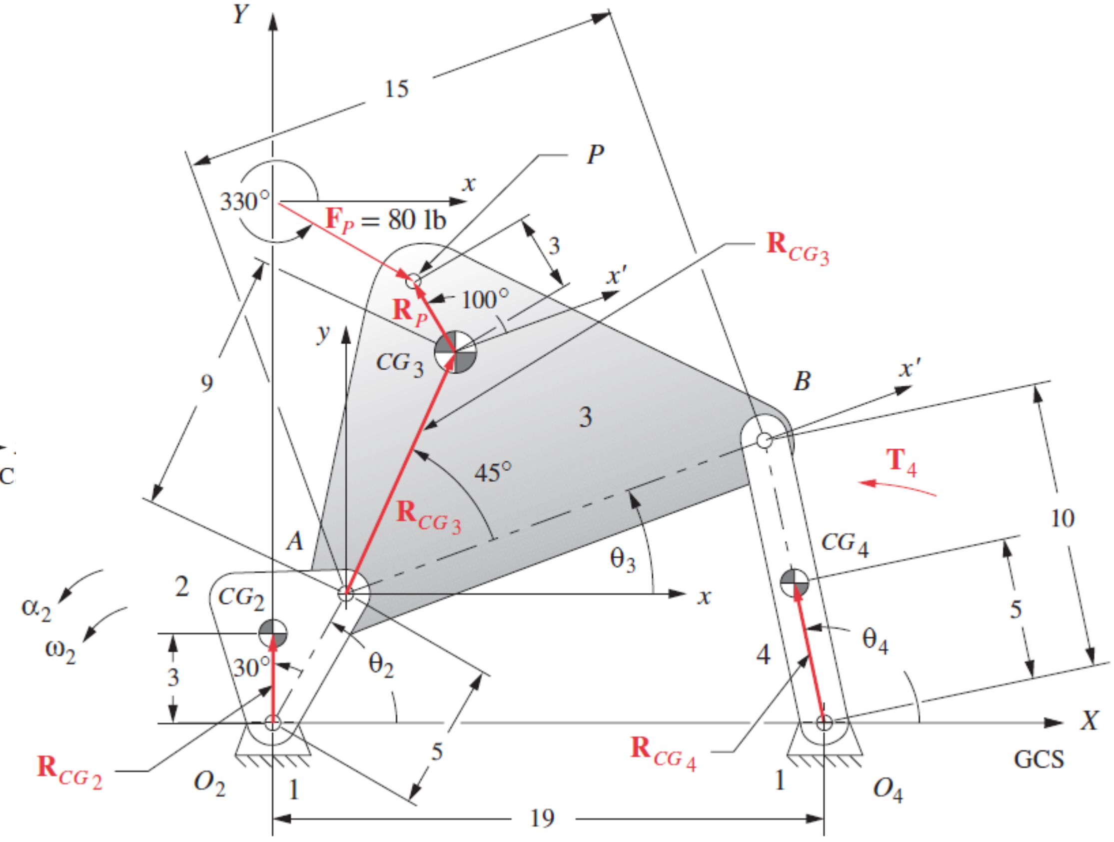
$$F_{12_x} = -57.67 \text{ lb}$$

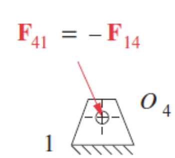
$$F_{12_y} = -9.39 \text{ lb}$$

$$T_{12} = 204.72 \text{ lb-in}$$

# – Fourbar linkage

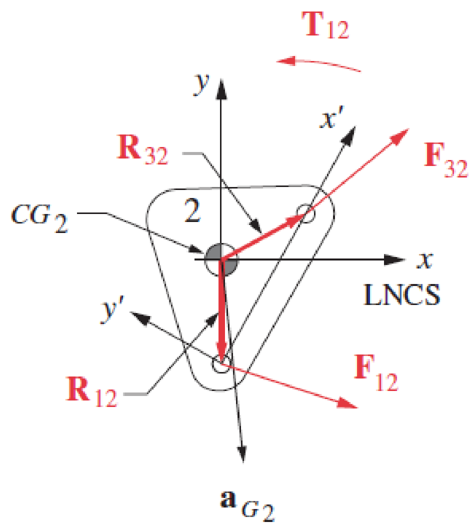



**LRCS**: local rotating axis coordinate system
**LNCS**: local nonrotating axis coordinate system
**GCS**: global coordinate system

# Link 2

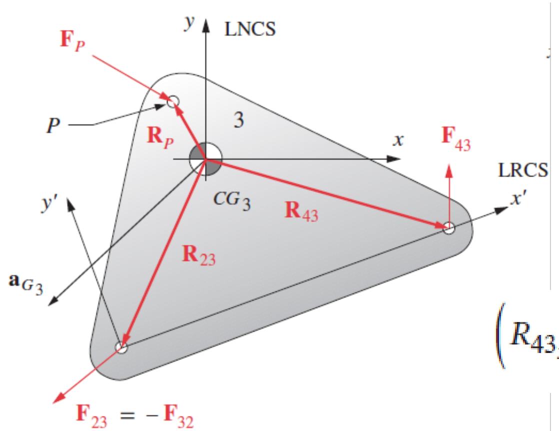



$$F_{12_x} + F_{32_x} = m_2 a_{G_{2x}}$$

$$F_{12_y} + F_{32_y} = m_2 a_{G_{2y}}$$

$$T_{12} + \left(R_{12_x} F_{12_y} - R_{12_y} F_{12_x}\right) + \left(R_{32_x} F_{32_y} - R_{32_y} F_{32_x}\right) = I_{G_2} \alpha_2$$

# Link 3

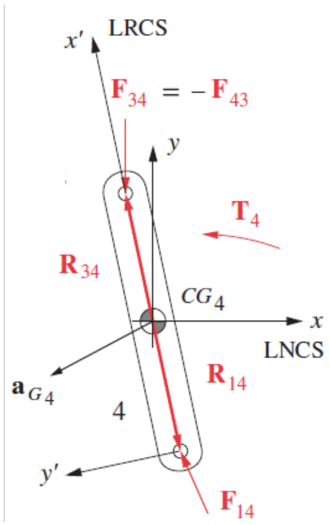



$$F_{43_x} - F_{32_x} + F_{P_x} = m_3 a_{G_{3x}}$$

$$F_{43_y} - F_{32_y} + F_{P_y} = m_3 a_{G_{3y}}$$

$$\left(R_{43_x} F_{43_y} - R_{43_y} F_{43_x}\right) - \left(R_{23_x} F_{32_y} - R_{23_y} F_{32_x}\right) + \left(R_{P_x} F_{P_y} - R_{P_y} F_{P_x}\right) = I_{G_3} \alpha_3$$

# Link 4




$$F_{14_x} - F_{43_x} = m_4 a_{G_{4x}}$$

$$F_{14_y} - F_{43_y} = m_4 a_{G_{4y}}$$

$$\left(R_{14_x} F_{14_y} - R_{14_y} F_{14_x}\right) - \left(R_{34_x} F_{43_y} - R_{34_y} F_{43_x}\right) + T_4 = I_{G_4} \alpha_4$$

# Matrix equation

$$\begin{bmatrix} 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ -R_{12_y} & R_{12_x} & -R_{32_y} & R_{32_x} & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & R_{23_y} & -R_{23_x} & -R_{43_y} & R_{43_x} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & R_{34_y} & -R_{34_x} & -R_{14_y} & R_{14_x} & 0 \end{bmatrix} \begin{bmatrix} F_{12_x} \\ F_{12_y} \\ F_{32_x} \\ F_{32_y} \\ F_{43_x} \\ F_{43_y} \\ F_{14_x} \\ F_{14_y} \\ T_{12} \end{bmatrix} = \begin{bmatrix} m_2 a_{G_{2x}} \\ m_2 a_{G_{2y}} \\ I_{G_2} \alpha_2 \\ m_3 a_{G_{3x}} - F_{P_x} \\ m_3 a_{G_{3y}} - F_{P_y} \\ I_{G_3} \alpha_3 - R_{P_x} F_{P_y} + R_{P_y} F_{P_x} \\ m_4 a_{G_{4x}} \\ m_4 a_{G_{4y}} \\ I_{G_4} \alpha_4 - T_4 \end{bmatrix}$$

# Example 2

The 5-in-long crank (link 2) shown weighs 1.5 lb. Its *CG* is at 3 in @ +30° from the line of centers (LRCS). Its mass moment of inertia about its *CG* is 0.4 lb-in-sec$^2$. Its kinematic data are:

| $\theta_2$ deg | $\omega_2$ rad/sec | $\alpha_2$ rad/sec$^2$ | $a_{G_2}$ in/sec$^2$ |
|---|---|---|---|
| 60 | 25 | –40 | 1878.84 @ 273.66° |

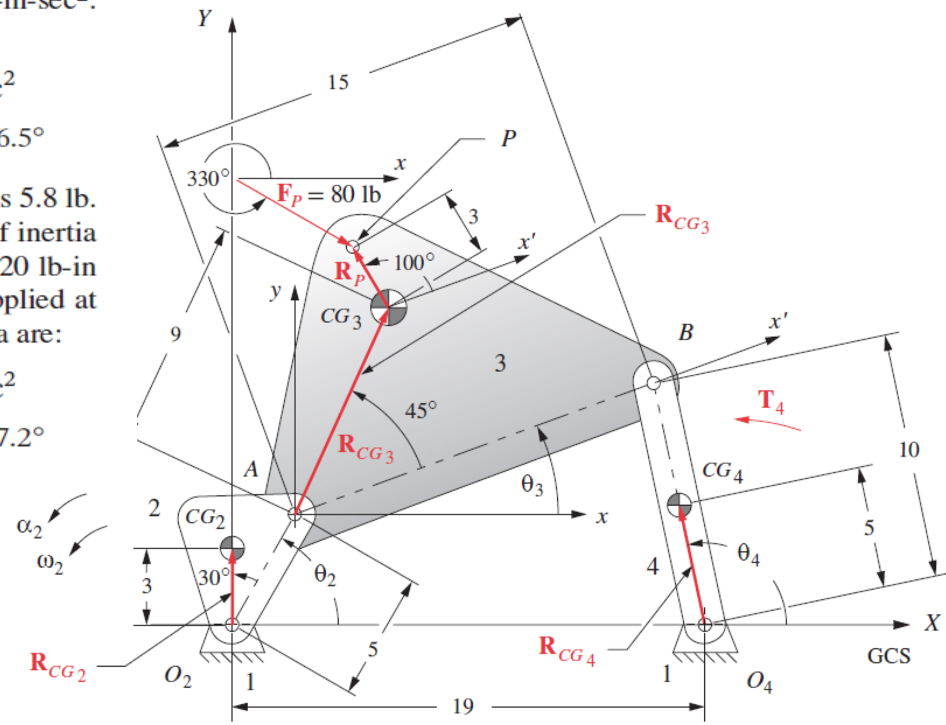
The coupler (link 3) is 15 in long and weighs 7.7 lb. Its *CG* is at 9 in @ 45° off the line of centers (LRCS). Its mass moment of inertia about its *CG* is 1.5 lb-in-sec$^2$. Its kinematic data are:

| $\theta_3$ deg | $\omega_3$ rad/sec | $\alpha_3$ rad/sec$^2$ | $a_{G_3}$ in/sec$^2$ |
|---|---|---|---|
| 20.92 | –5.87 | 120.9 | 3646.1 @ 226.5° |

The ground link is 19 in long. The rocker (link 4) is 10 in long and weighs 5.8 lb. Its *CG* is at 5 in @ 0° on the line of centers (LRCS). Its mass moment of inertia about its *CG* is 0.8 lb-in-sec$^2$. There is an external torque on link 4 of 120 lb-in (GCS). An external force of 80 lb @ 330° acts on link 3 in the GCS, applied at point *P* at 3 in @ 100° from the *CG* of link 3 (LRCS). The kinematic data are:

| $\theta_4$ deg | $\omega_4$ rad/sec | $\alpha_4$ rad/sec$^2$ | $a_{G_4}$ in/sec$^2$ |
|---|---|---|---|
| 104.41 | 7.93 | 276.29 | 1416.8 @ 207.2° |

***Find:*** Forces **F**12, **F**32, **F**43, and **F**14 at the joints and the driving torque **T**12 needed to maintain motion with the given acceleration for this instantaneous position of the link.



# Solution

$$m_2 = \frac{weight}{g} = \frac{1.5\ \text{lb}}{386\ \text{in/sec}^2} = 0.004\ \ \text{blob}$$

$$m_3 = \frac{weight}{g} = \frac{7.7\ \text{lb}}{386\ \text{in/sec}^2} = 0.020\ \ \text{blob}$$

$$m_4 = \frac{weight}{g} = \frac{5.8\ \text{lb}}{386\ \text{in/sec}^2} = 0.015\ \ \text{blob}$$

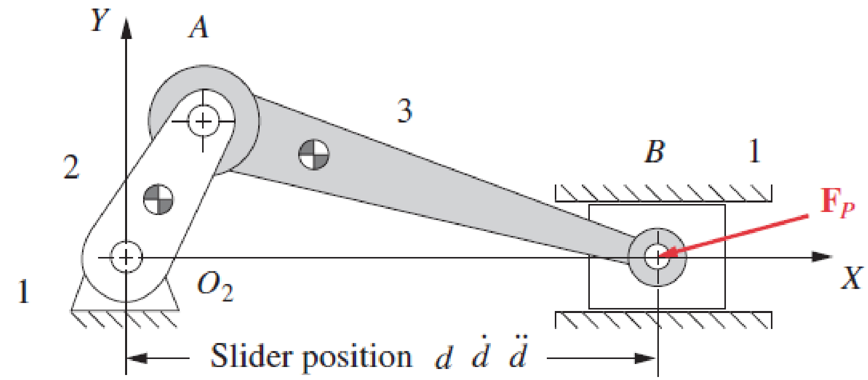
$$\begin{aligned}
\mathbf{R}_{12} &= 3.00 \ @ \angle\ 270.00°; & R_{12_x} &= 0.000, & R_{12_y} &= -3\\
\mathbf{R}_{32} &= 2.83 \ @ \angle\ 28.00°; & R_{32_x} &= 2.500, & R_{32_y} &= 1.333\\
\mathbf{R}_{23} &= 9.00 \ @ \angle\ 245.92°; & R_{23_x} &= -3.672, & R_{23_y} &= -8.217\\
\mathbf{R}_{43} &= 10.72 \ @ \angle\ -15.46°; & R_{43_x} &= 10.332, & R_{43_y} &= -2.858\\
\mathbf{R}_{34} &= 5.00 \ @ \angle\ 104.41°; & R_{34_x} &= -1.244, & R_{34_y} &= 4.843\\
\mathbf{R}_{14} &= 5.00 \ @ \angle\ 284.41°; & R_{14_x} &= 1.244, & R_{14_y} &= -4.843\\
\mathbf{R}_{P} &= 3.00 \ @ \angle\ 120.92°; & R_{P_x} &= -1.542, & R_{P_y} &= 2.574\\
\mathbf{a}_{G_2} &= 1878.84\ @ \angle 273.66°; & a_{G_{2x}} &= 119.94, & a_{G_{2y}} &= -1875.01\\
\mathbf{a}_{G_3} &= 3646.10\ @ \angle 226.51°; & a_{G_{3x}} &= -2509.35, & a_{G_{3y}} &= -2645.23\\
\mathbf{a}_{G_4} &= 1416.80\ @ \angle 207.24°; & a_{G_{4x}} &= -1259.67, & a_{G_{4y}} &= -648.50
\end{aligned}$$

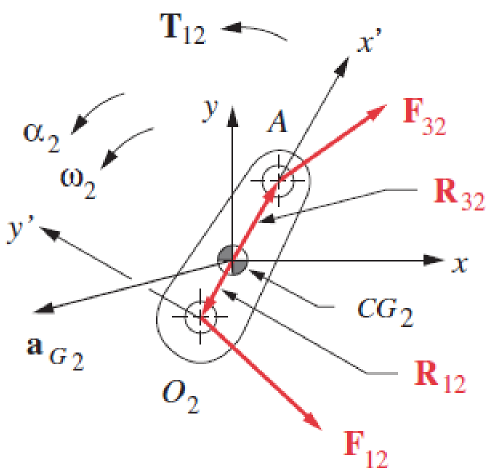
$$\mathbf{F}_{P3} = 80 \ \ @ \ \angle\ 330°; \qquad F_{P3_x} = 69.28, \qquad F_{P3_y} = -40.00$$

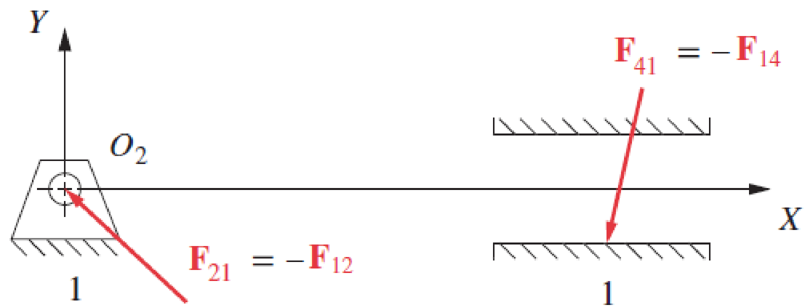
$$\begin{bmatrix} F_{12_x} \\ F_{12_y} \\ F_{32_x} \\ F_{32_y} \\ F_{43_x} \\ F_{43_y} \\ F_{14_x} \\ F_{14_y} \\ T_{12} \end{bmatrix} = \begin{bmatrix} -117.65 \\ -107.84 \\ 118.13 \\ 100.34 \\ -1.34 \\ 87.43 \\ -20.23 \\ 77.71 \\ 243.23 \end{bmatrix}$$

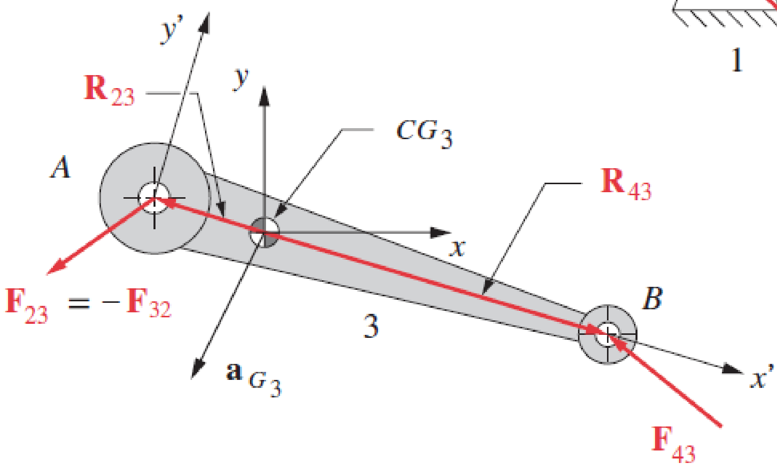
Substitute the above values into the matrix equation in the last slide and solve it:

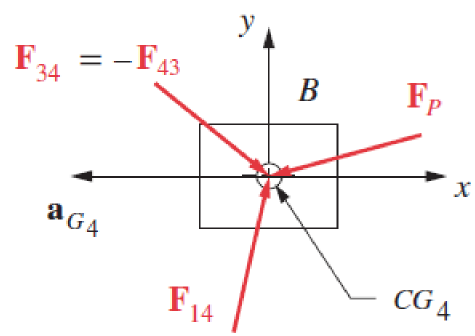
– **Crank Slider**



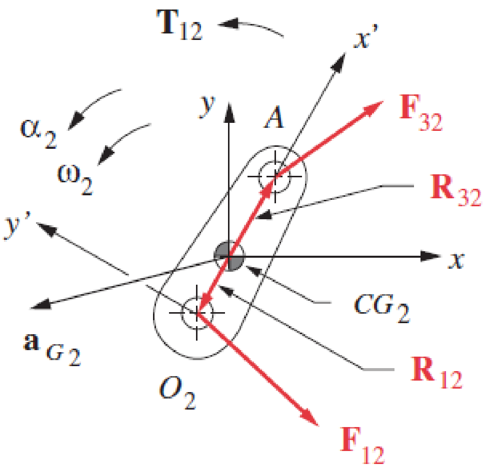



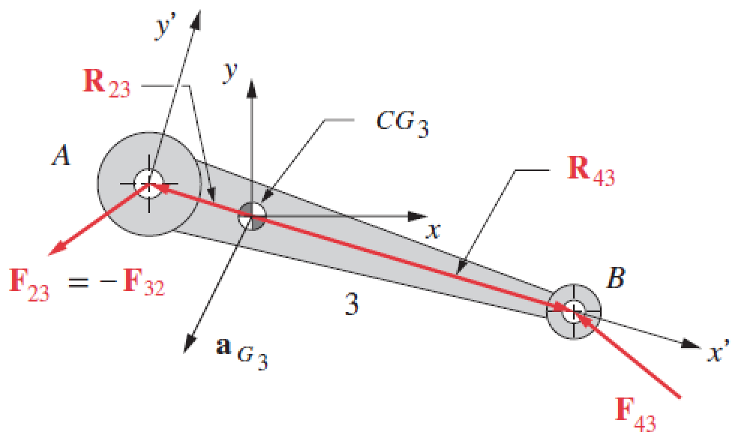



# Link 2



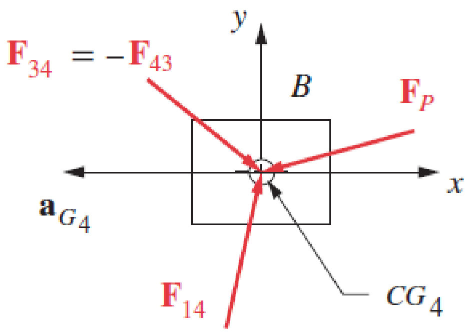

$$F_{12_x} + F_{32_x} = m_2 a_{G_{2x}}$$
$$F_{12_y} + F_{32_y} = m_2 a_{G_{2y}}$$
$$T_{12} + \left(R_{12_x} F_{12_y} - R_{12_y} F_{12_x}\right) + \left(R_{32_x} F_{32_y} - R_{32_y} F_{32_x}\right) = I_{G_2} \alpha_2$$

# Link 3




$$F_{43_x} - F_{32_x} = m_3 a_{G_{3x}}$$
$$F_{43_y} - F_{32_y} = m_3 a_{G_{3y}}$$
$$\left(R_{43_x} F_{43_y} - R_{43_y} F_{43_x}\right) - \left(R_{23_x} F_{32_y} - R_{23_y} F_{32_x}\right) = I_{G_3} \alpha_3$$

# Link 4




$$F_{14_x} - F_{43_x} + F_{P_x} = m_4 a_{G_{4x}}$$
$$F_{14_y} - F_{43_y} + F_{P_y} = m_4 a_{G_{4y}}$$
$$F_{14_x} = -\mu\, SGN\left(\dot{d}\right)\left|F_{14_y}\right|$$

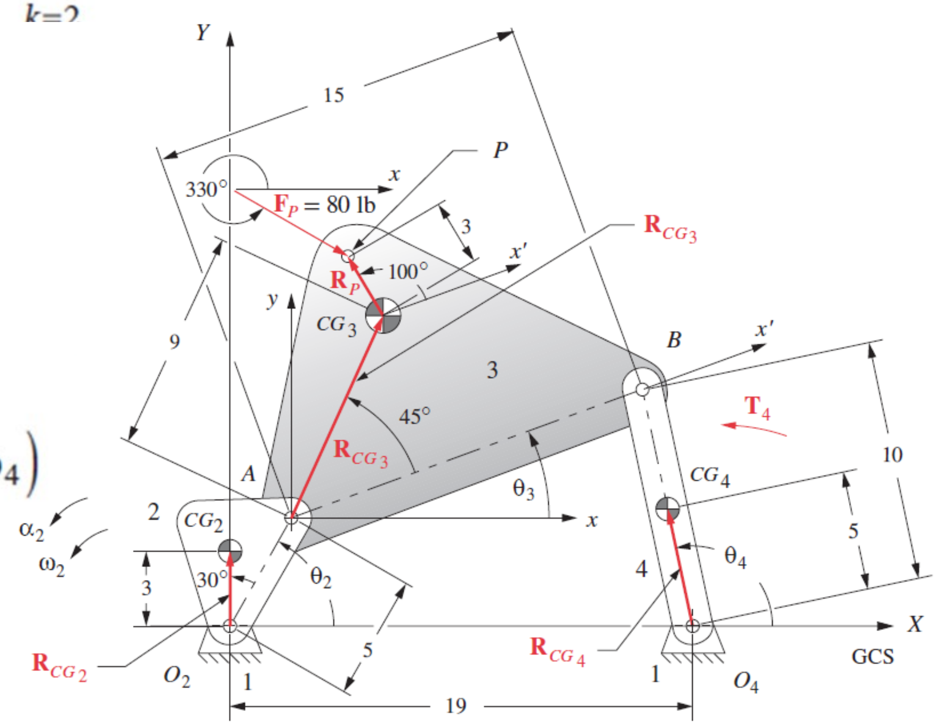
# Matrix equation

$$\begin{bmatrix} 1 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ -R_{12_y} & R_{12_x} & -R_{32_y} & R_{32_x} & 0 & 0 & 0 & 1 \\ 0 & 0 & -1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 1 & 0 & 0 \\ 0 & 0 & R_{23_y} & -R_{23_x} & -R_{43_y} & R_{43_x} & 0 & 0 \\ 0 & 0 & 0 & 0 & -1 & 0 & -\mu SGN\left(\dot{d}\right) & 0 \\ 0 & 0 & 0 & 0 & 0 & -1 & 1 & 0 \end{bmatrix} \begin{bmatrix} F_{12_x} \\ F_{12_y} \\ F_{32_x} \\ F_{32_y} \\ F_{43_x} \\ F_{43_y} \\ F_{14_y} \\ T_{12} \end{bmatrix} = \begin{bmatrix} m_2 a_{G_{2x}} \\ m_2 a_{G_{2y}} \\ I_{G_2}\alpha_2 \\ m_3 a_{G_{3x}} \\ m_3 a_{G_{3y}} \\ I_{G_3}\alpha_3 \\ m_4 a_{G_{4x}} - F_{P_x} \\ -F_{P_y} \end{bmatrix}$$

- **Virtual Work / Energy Method (*to* *<u>solve externally applied force</u>* *only*)**

$$\sum_{k=2}^{n}\underline{\mathbf{F}_k}\cdot\mathbf{v}_k+\sum_{k=2}^{n}\underline{\mathbf{T}_k}\cdot\boldsymbol{\omega}_k-\sum_{k=2}^{n}m_k\mathbf{a}_k\cdot\mathbf{v}_k-\sum_{k=2}^{n}I_k\alpha_k\cdot\boldsymbol{\omega}_k=0$$

$$\left(\mathbf{F}_{P_3}\cdot\mathbf{v}_{P_3}+\overset{\boxed{0}}{\cancel{\mathbf{F}_{P_4}}}\cdot\mathbf{v}_{P_4}\right)+\left(\mathbf{T}_{12}\cdot\boldsymbol{\omega}_2+\overset{\boxed{0}}{\cancel{\mathbf{T}_3}}\cdot\boldsymbol{\omega}_3+\mathbf{T}_4\cdot\boldsymbol{\omega}_4\right)$$
$$=\left(m_2\mathbf{a}_{G_2}\cdot\mathbf{v}_{G_2}+m_3\mathbf{a}_{G_3}\cdot\mathbf{v}_{G_3}+m_4\mathbf{a}_{G_4}\cdot\mathbf{v}_{G_4}\right)$$
$$+\left(I_{G_2}\alpha_2\cdot\boldsymbol{\omega}_2+I_{G_3}\alpha_3\cdot\boldsymbol{\omega}_3+I_{G_4}\alpha_4\cdot\boldsymbol{\omega}_4\right)$$




$$\left(F_{P_{3x}}V_{P_{3x}}+F_{P_{3y}}V_{P_{3y}}\right)+\left(\overset{\boxed{0}}{\cancel{F_{P_{4x}}}}V_{P_{4x}}+\overset{\boxed{0}}{\cancel{F_{P_{4y}}}}V_{P_{4y}}\right)+\left(T_{12}\omega_2+T_3\omega_3+T_4\omega_4\right)$$
$$=m_2\left(a_{G_{2x}}V_{G_{2x}}+a_{G_{2y}}V_{G_{2y}}\right)+m_3\left(a_{G_{3x}}V_{G_{3x}}+a_{G_{3y}}V_{G_{3y}}\right)$$
$$+m_4\left(a_{G_{4x}}V_{G_{4x}}+a_{G_{4y}}V_{G_{4y}}\right)+\left(I_{G_2}\alpha_2\omega_2+I_{G_3}\alpha_3\omega_3+I_{G_4}\alpha_4\omega_4\right)$$

# Example 3

The 5-in-long crank (link 2) shown weighs 1.5 lb. Its *CG* is at 3 in at +30° from the line of centers. Its mass moment of inertia about its *CG* is 0.4 lb-in-sec$^2$. Its kinematic data are:

| $\theta_2$ deg | $\omega_2$ rad/sec | $\alpha_2$ rad/sec$^2$ | $V_{G_2}$ in/sec |
|---|---|---|---|
| 60 | 25 | –40 | 75 @ 180° |

The coupler (link 3) is 15 in long and weighs 7.7 lb. Its *CG* is at 9 in at 45° off the line of centers. Its mass moment of inertia about its *CG* is 1.5 lb-in-sec$^2$. Its kinematic data are:

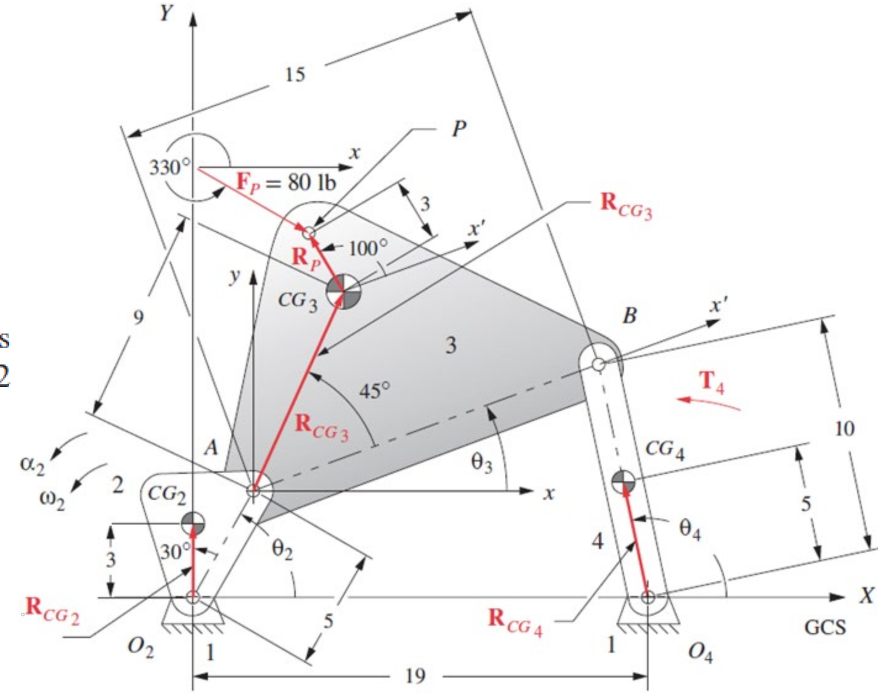
| $\theta_3$ deg | $\omega_3$ rad/sec | $\alpha_3$ rad/sec$^2$ | $V_{G_3}$ in/sec |
|---|---|---|---|
| 20.92 | –5.87 | 120.9 | 72.66 @ 145.7° |

There is an external force on link 3 of 80 lb at 330°, applied at point *P* which is located 3 in @ 100° from the *CG* of link 3. The linear velocity of that point is 67.2 in/sec at 131.94°.

The rocker (link 4) is 10-in long and weighs 5.8 lb. Its *CG* is at 5 in at 0° off the line of centers. Its mass moment of inertia about its *CG* is 0.8 lb-in-sec$^2$. Its data are:

| $\theta_4$ deg | $\omega_4$ rad/sec | $\alpha_4$ rad/sec$^2$ | $V_{G_4}$ in/sec |
|---|---|---|---|
| 104.41 | 7.93 | 276.29 | 39.66 @ 194.41° |

There is an external torque on link 4 of 120 lb-in. The ground link is 19-in long.




***Find:*** The driving torque T12 needed to maintain motion with the given acceleration for this instantaneous position of the link.

# Solution

$$m_2 = \frac{weight}{g} = \frac{1.5 \text{ lb}}{386 \text{ in/sec}^2} = 0.004 \text{ blob}$$

$$m_3 = \frac{weight}{g} = \frac{7.7 \text{ lb}}{386 \text{ in/sec}^2} = 0.020 \text{ blob}$$

$$m_4 = \frac{weight}{g} = \frac{5.8 \text{ lb}}{386 \text{ in/sec}^2} = 0.015 \text{ blob}$$

$$\mathbf{a}_{G_2} = 1878.84 \;@\; \angle 273.66°; \quad a_{G_{2x}} = 119.94, \quad a_{G_{2y}} = -1875.01$$

$$\mathbf{a}_{G_3} = 3646.10 \;@\; \angle 226.51°; \quad a_{G_{3x}} = -2509.35, \quad a_{G_{3y}} = -2645.23$$

$$\mathbf{a}_{G_4} = 1416.80 \;@\; \angle 207.24°; \quad a_{G_{4x}} = -1259.67, \quad a_{G_{4y}} = -648.50$$

$$\mathbf{V}_{G_2} = 75.00 \;@\angle\; 180.00°; \quad V_{G_{2x}} = -75.00, \quad V_{G_{2y}} = 0$$

$$\mathbf{V}_{G_3} = 72.66 \;@\angle\; 145.70°; \quad V_{G_{3x}} = -60.02, \quad V_{G_{3y}} = 40.95$$

$$\mathbf{V}_{G_4} = 39.66 \;@\angle\; 194.41°; \quad V_{G_{4x}} = -38.41, \quad V_{G_{4y}} = -9.87$$

$$\mathbf{V}_{P_3} = 67.20 \;@\angle\; 131.94°; \quad V_{P_{3x}} = -44.91, \quad V_{P_{3y}} = 49.99$$

$$\mathbf{F}_{P3} = 80 \;@\; \angle\; 330°; \quad F_{P3_x} = 69.28, \quad F_{P3_y} = -40.00$$

Applying

$$\left(F_{P_{3x}} V_{P_{3x}} + F_{P_{3y}} V_{P_{3y}}\right) + \left(\cancel{F_{P_{4x}}} V_{P_{4x}} + \cancel{F_{P_{4y}}} V_{P_{4y}}\right) + \left(T_{12}\omega_2 + T_3\omega_3 + T_4\omega_4\right)$$
$$= m_2\left(a_{G_{2x}} V_{G_{2x}} + a_{G_{2y}} V_{G_{2y}}\right) + m_3\left(a_{G_{3x}} V_{G_{3x}} + a_{G_{3y}} V_{G_{3y}}\right)$$
$$+ m_4\left(a_{G_{4x}} V_{G_{4x}} + a_{G_{4y}} V_{G_{4y}}\right) + \left(I_{G_2}\alpha_2\omega_2 + I_{G_3}\alpha_3\omega_3 + I_{G_4}\alpha_4\omega_4\right)$$

$$0.004.28)(-44.91) + (-40)(49.99)] + [0] + [25T_{12} + (0) + (120)(7.93)]$$
$$= 0.004\left[(119.94)(-75) + (-1875.01)(0)\right)$$
$$+ 0.020\left[(-2509.35)(-60.02) + (-2645.23)(40.95)\right]$$
$$+ 0.015\left[(-1259.67)(-38.41) + (-648.50)(-9.87)\right]$$
$$+ \left[(0.4)(-40)(25) + (1.5)(120.9)(-5.87) + (0.8)(276.29)(7.93)\right]$$

$$\mathbf{T}_{12} = 243.2\ \hat{\mathbf{k}}$$

# • Shaking force and shaking moment of force

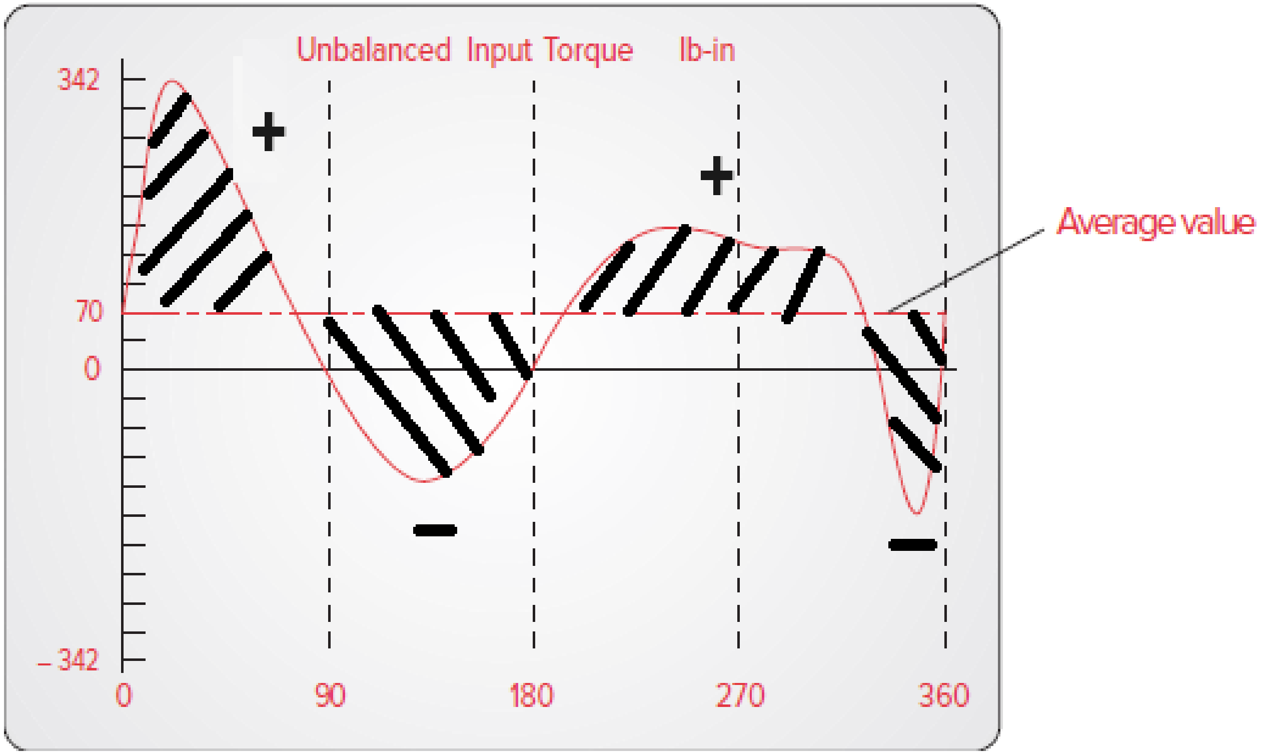
**Shaking force:** the sum of the forces acting on the ground

$$\mathbf{F}_s = \mathbf{F}_{21} + \mathbf{F}_{41}$$

**shaking moment of force:** the reaction moment of force felt by the ground

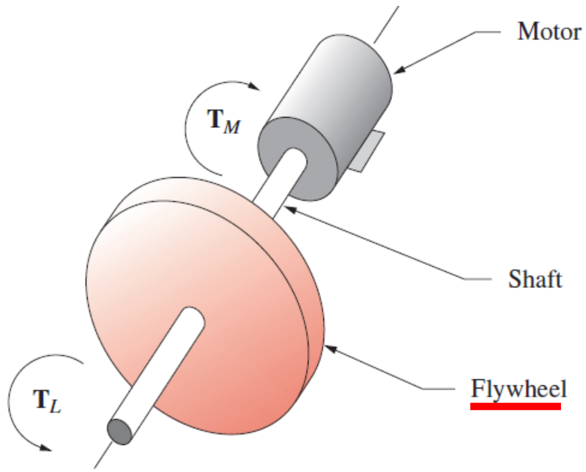
$$\mathbf{M}_s = \mathbf{T}_{21} + \left(\mathbf{R}_1 \times \mathbf{F}_{41}\right)$$

- **Input torque smoothing – flywheels (*passive torque control*)**




Input torque curve for an unbalanced crank-rocker fourbar linkage

(Drive crank speed: *50 rad/sec.*)




Kinetic energy storage

$$T_M - T_L = I\alpha$$

$T_M = T_{avg}$ (*constant; free of oscillations desired*)

$$\alpha = \frac{d\omega}{dt} = \frac{d\omega}{dt}\left(\frac{d\theta}{d\theta}\right) = \omega\frac{d\omega}{d\theta}$$

$$\left(T_{avg} - T_L\right)d\theta = I\omega\, d\omega$$

$$\int_{\theta@\omega_{min}}^{\theta@\omega_{max}} \left(T_{avg} - T_L\right)d\theta = \frac{1}{2}I\left(\omega_{max}^2 - \omega_{min}^2\right) = E$$

**Energy**

- Minimize the speed (torque) variation by providing a flywheel with a sufficiently large *moment of inertia (I)*.
- The *integration is done with respect to the average line of the torque function; aims to make* the sum of positive areas above an average line equal to the sum of negative areas below that line

## – Sizing the flywheel

$$E = \frac{1}{2} I_s \left( \omega_{max}^2 - \omega_{min}^2 \right)$$

$I_s$: *moment of inertia of the system including the flywheel*

*Fluctuation*: $Fl = \omega_{max} - \omega_{min}$

*Coefficient of fluctuation:* $k = \dfrac{\omega_{\max} - \omega_{\min}}{\omega_{avg}}$

$$E = \frac{1}{2} I_s \left( 2\omega_{avg} \right)\left( k\,\omega_{avg} \right)$$

$$I_s = \frac{E}{k\,\omega_{avg}^2}$$

$I_f = I_s$ *minus (-) the moment of inertia of other parts around the shaft*

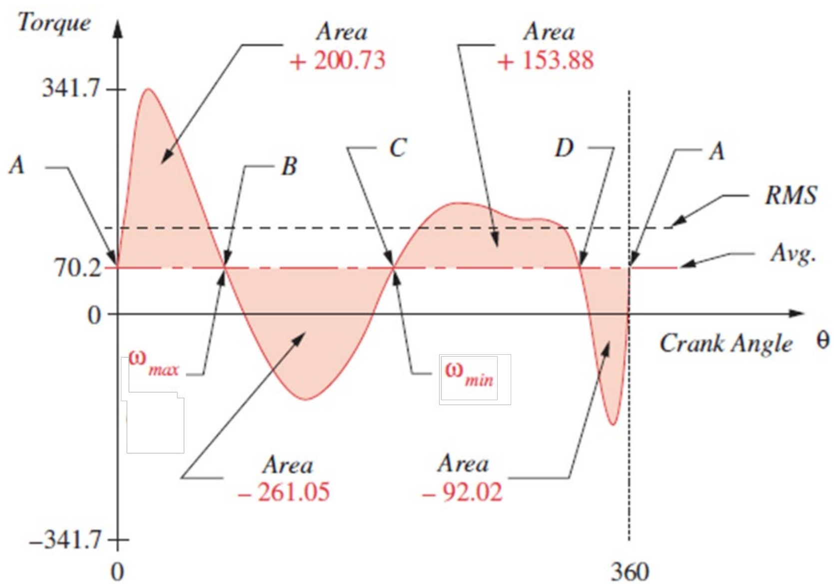
$I_f = mr^2 \Rightarrow r$ (*Radius of the wheel*)

# Example 4

Determining the Energy Variation in a Torque-Time Function.

***Given:*** An input torque-time function which varies over its cycle. Figure 11-11 shows the
input torque curve from Figure 11-8. The torque is varying during the 360° cycle about its average value.

***Find:*** The total energy variation over one cycle.



# Solution

| From | Area = $E$ | Accum. Sum = $E$ | |
|---|---|---|---|
| $A$ to $B$ | +200.73 | +200.73 | $\omega_{max}$ @$B$ |
| $B$ to $C$ | –261.05 | –60.32 | $\omega_{min}$ @$C$ |
| $C$ to $D$ | +153.88 | +93.56 | |
| $D$ to $A$ | –92.02 | +1.54 | |
| | Total Energy | $= E @ \omega_{max} - E @ \omega_{min}$ | |
| | | $= (+200.73) - (-60.32) = 261.05$ in-lb | |

- The maximum shaft speed occurs at the end of the largest accumulated positive energy pulse (+200.73 in-lb) to accelerate the shaft.

- The minimum shaft speed occurs at the end of the largest accumulated negative energy pulse (–60.32 in-lb) to decelerate the shaft.

Input torque curve for the linkage in Figure 11-8 after smoothing with a flywheel